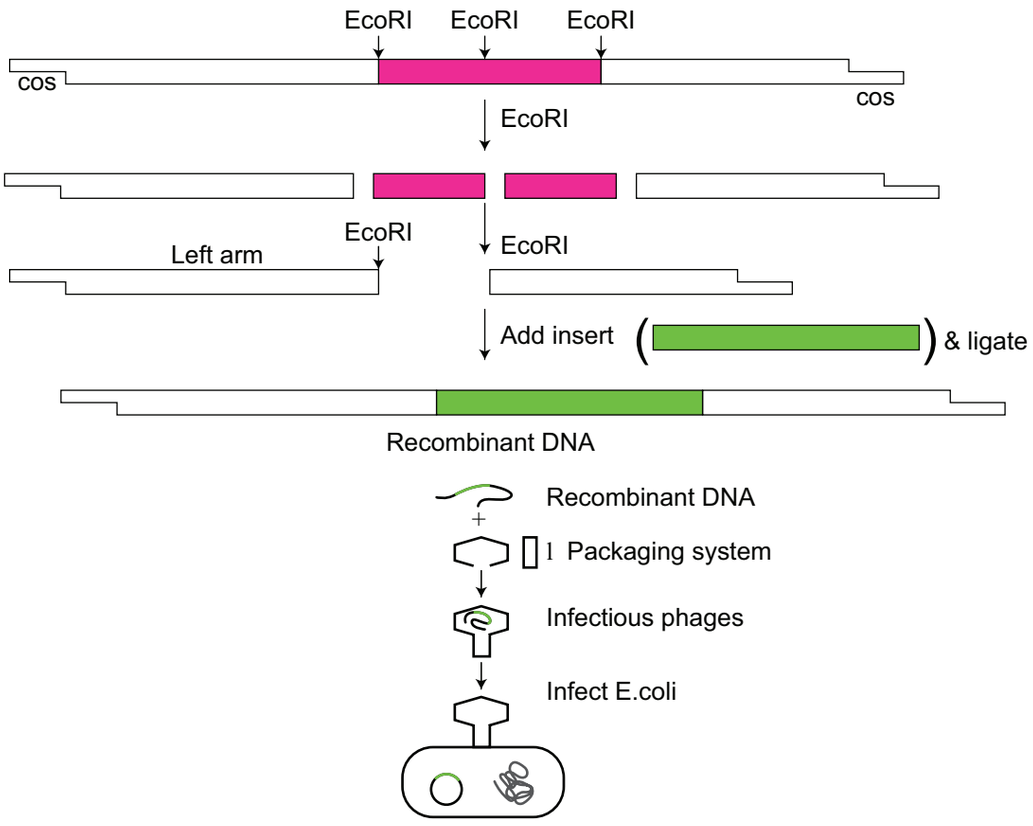


Figure 12.28: Cloning using a λ phage

Phagemids: Phagemids are the combination of plasmid and phage and can function as either plasmid or phage. Since they posses functional origins of replication of both plasmid and phage λ they can be propagated (as plasmid or phage) in appropriate E.coli.

Artificial chromosome Vectors: Artificial chromosomes are cloning vectors that can accommodate very large pieces of DNA, producing recombinant DNA molecules resembling small chromosomes.

1. Yeast artificial chromosome YAC is a synthetic DNA that can accept large fragments of foreign DNA between 0.2 to 2.0 Mb (particularly human DNA). In addition to origin of replication sequence and selectable markers they possess centromeric and telomeric regions, and therefore the recombinant DNA can be maintained like a yeast chromosome.
2. (BACs) Bacterial Artificial chromosomes: BACS can accept DNA inserts around 300 kb.

A major part of the sequencing of human genome has been accomplished by using a library of BAC recombinant. BACS are vectors containing the origin replication of a natural plasmid called the F factor, a MCS, a selectable marker and often some other features.

Plasmid shuttle Vectors: The plasmid vectors that are specifically designed to replicate in two or more different host organisms(say in E.coli and yeast) are referred to as shuttle vectors. The origins of replication for two hosts are combined in one plasmid.

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